

Indian Institute of Technology Bhilai

(An Institute of National Importance under Ministry of Education, Govt. of India)

Kutlabhata, Durg – 491002, Chhattisgarh, India



Dr. Jose Immanuel R.

Outstanding Young Faculty Chair

Assistant Professor, Department of Mechanical Engineering

31 July 2024

To,
The Chairman of selection panel,
HYUNDAI HOPE SCHOLARSHIP PROGRAM.

I take pleasure in writing this letter of recommendation for the team, headed by **Mr. Mohit**, with team members **Mr. Gourav** and **Mr. Maneesh**, for Hyundai Hope Scholarship program 2024. The team is submitting a proposal titled “AI Integrated Machine vision technology for equipment condition monitoring and feed control in Indian Industries with a use case from our region”. The students are pursuing their BTech program and are currently in third year of their four-year BTech program. I personally know the team since 2024 March, when they started to work on the said problem.

During my first interaction with the team, I could see the curiosity in their eyes when they came up with an idea of working on industrial relevant problems. They went all around the industrial area of the city, interacting with engineers of various industries to understand what is the current practice in industries on process optimization, so that they can work on a technological solution. They have collated the information gathered, and then came up with this idea of developing a data-driven condition monitoring and process control approach.

The idea of theirs is novel to the sector in which they are projecting and shall create a large positive impact on their productivity and subsequently finance. I see that the team is well equipped with a diversified expertise amongst the members. **Gourav** and **Mohit** have strong fundamentals on Mechatronic systems development, acquiring and processing of sensors data, design of actuations systems, etc. They have worked on short projects in developing robotic systems for industrial fault analysis, automobile autonomy, etc. **Maneesh** has a sound background on Data science, with hands-on experience on AI tools for image processing. He has worked on software development projects in the past which will help the team in scaling up the projected solutions in future as well.

The students are self-motivated and have come forward on their own to work on this problem. They have already spent a lot of time, effort and money to bring the project to the shape it is currently. They need some motivation along with financial support to carry the work forward, and I believe the Hyundai scholarship would give them the required motivation and assistance.

With this, I give my strong recommendation for the team and my best wishes for their success.


Jose Immanuel R.